

Pedro Sandoval Segura

CONTACT INFORMATION	3116 Brendan Iribe Center 8125 Paint Branch Drive College Park, MD 20742	(415) 324 9232 psando@umd.edu http://cs.umd.edu/~psando
RESEARCH INTERESTS	I am broadly interested in machine learning and deep learning. I believe that better understanding what and how models learn will lead to more efficient AI methods.	
EDUCATION	University of Maryland , College Park, MD Ph.D., Computer Science Advised by <i>Prof. Tom Goldstein</i> and <i>Prof. David Jacobs</i> <i>National Defense Science and Engineering Graduate Fellow</i> Harvey Mudd College , Claremont, CA B.S., Computer Science and Mathematics <i>Graduated with High Distinction</i>	
		Expected 2026 May 2019
RESEARCH EXPERIENCE	Apple , Cupertino, CA <i>Deep Learning Research Intern</i> - Investigated application of Encoder-Decoder Transformers for use in vectorization of geospatial data - Presented to Apple Maps leadership, where results were described as a “breakthrough” - Working with Dr. Yonghong Wang Amazon Lab126 , Sunnyvale, CA <i>Applied Scientist Intern</i> - Analyzed approaches for training NeRF models and proposed a method for consistent depth estimation - Working with Dr. Chi Liu and Dr. Albert Chen U.S. Naval Research Laboratory , Washington, D.C. - Investigating adversarial examples, robustness, and interpretability in meta-learning approaches for few-shot learning - Submitted findings to 5th Workshop on Meta-Learning at NeurIPS 2021 - Working with Dr. Ed Lawson	
		Summer 2023 Summer 2022 Summer 2021
INDUSTRY EXPERIENCE	Facebook, Inc. , Seattle, WA <i>Software Engineering Intern</i> - Implemented and monitored new share flow functionality for encrypted Messenger threads, allowing users to forward text, sticker, photo, audio, and video securely - Received a full-time offer at the conclusion of internship Facebook, Inc. , Menlo Park, CA <i>Software Engineering Intern</i> - Designed and built a new Messenger Groups Tab approvals surface, enabling users to accept join requests across multiple group threads - Oversaw, implemented, and ran an A/B test interleaving active groups in the Active Tab which drove topline metrics such as group sends and group creates	
		Summer 2018 Summer 2017

Facebook, Inc., Menlo Park, CA
Facebook University for Engineering Intern

Summer 2016

- Organized engineering tasks, drafted feature ideas, and collaborated with a team of 3 to build InSync, an iOS app which synchronizes music on multiple devices

PUBLICATIONS

10. **Sandoval-Segura**, Wang, Panda, Goldblum, Basri, Goldstein, Jacobs. “Identifying and Evaluating Inactive Heads in Pretrained LLMs”. *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026. (28% acceptance rate)
9. Kim, **Sandoval-Segura**. “Learning from Convolution-based Unlearnable Datasets”. *NeurIPS Workshop on New Frontiers in Adversarial Machine Learning (Adv-ML)*, 2024.
8. Singla, **Sandoval-Segura**, Goldblum, Geiping, Goldstein. “A Simple and Efficient Baseline for Data Attribution on Images”. *NeurIPS Workshop on Attributing Model Behavior at Scale (ATTRIB)*, 2023.
7. **Sandoval-Segura**, Singla, Geiping, Goldblum, Goldstein. “What Can We Learn from Unlearnable Datasets?”. *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 2023. (26.1% acceptance rate)
6. **Sandoval-Segura**, Singla, Geiping, Goldblum, Goldstein, Jacobs. “Autoregressive Perturbations for Data Poisoning”. *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2022. (25.6% acceptance rate)
5. **Sandoval-Segura**, Singla, Fowl, Geiping, Goldblum, Jacobs, Goldstein. “Poisons that are learned faster are more effective”. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2022.
4. Bashir, Montañez, Sehra, **Sandoval-Segura**, Lauw. “An Information-Theoretic Perspective on Overfitting and Underfitting”. In *Australasian Joint Conference on Artificial Intelligence (AJCAI)*, 2020.
3. **Sandoval-Segura**, Lauw, Bashir, Shah, Sehra, Macias, Montañez. “The Labeling Distribution Matrix (LDM): A Tool for Estimating Machine Learning Algorithm Capacity”. *12th International Conference on Agents and Artificial Intelligence (ICAART)*, 2020.
2. Drissi, **Sandoval**, Ojha, Medero. “Harvey Mudd College at SemEval-2019 Task 4: The Clint-Buchanan Hyperpartisan News Detector”. In *Proceedings of The 13th International Workshop on Semantic Evaluation (SemEval)*, 2019.
1. Drissi, Watkins, Khant, Ojha, **Sandoval**, Segev, Weiner, Keller. “Programming Language Translation using a Grammar-Driven Tree-to-Tree Model”. *ICML Workshop on Neural Abstract Machines and Program Induction v2 (NAMPI)*, 2018.

HONORS AND AWARDS

National Defense Science and Engineering Graduate Fellowship	2023
Amazon Lab126 Diversity in Robotics and AI Fellowship	2021 - 2022
NeurIPS Scholar Award (\$1,625)	2022
Google CS Research Mentorship Program (CSRMP)	2021
Richard Tapia Conference Scholarship	2020
CRA-WP Grad Cohort for URMD	2020
UMD International Conference Student Support Award (\$500)	2020
UMD Dean’s Fellowship Program (\$5,000)	2019 - 2020
ARCS Scholarship, Los Angeles Chapter	2016 - 2019
Students Rising Above Scholarship	2015 - 2019

TEACHING EXPERIENCE	CMSC421: Introduction to Artificial Intelligence Graduate Teaching Assistant, University of Maryland	Spring 2021
	CMSC421: Introduction to Artificial Intelligence Graduate Teaching Assistant, University of Maryland	Fall 2020
	CMSC436: Programming Handheld Systems Graduate Teaching Assistant, University of Maryland	Fall 2019
	CS81: Computability and Logic Teaching Assistant, Harvey Mudd College	Spring 2019
	MATH187: Operations Research Grader, Harvey Mudd College	Spring 2019
LEADERSHIP AND OUTREACH	Emergency Reviewer for CVPR 2025	Spring 2025
	Program Committee for AAAI 2025	Fall 2024
	Program Committee for ATTRIB Workshop at NeurIPS 2024	Fall 2024
	Reviewer for ECCV 2024 + Outstanding Reviewer Award	Spring 2024
	Reviewer for CVPR 2024	Spring 2024
	Reviewer for NeurIPS 2023 and BUGS Workshop	Fall 2023
	Volunteer for NeurIPS 2023	Fall 2023
	Workshop Speaker at 17th Annual Winter Student Leadership Retreat (WSLR)	Spring 2023
	Reviewer for 1st International Conference on Automated ML	Spring 2022
	Reviewer for 5th Workshop on Meta-Learning at NeurIPS	Fall 2021
	CS Department M.S. and Ph.D. Admissions Committee	Spring 2020
	UMD Graduate Student Government Representative	Fall 2019
TECHNICAL BACKGROUND	JumpStart Computing Workshop, Iribe Initiative for Inclusion & Diversity in Computing	Fall 2019
	Languages: Python, Java, C, C++, Objective-C, Swift, Go, Haskell, AMPL, Arduino Software: PyTorch, Git, GitHub, Mercurial, Mathematica, MATLAB, Xcode	
PERSONAL INFORMATION	Citizenship: United States Language: English, Spanish	